

AI-Powered Drone System for Coconut Disease Detection, Real-time Monitoring and Intervention

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Project Proposal Report

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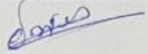
B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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Abstract

The coconut industry is a cornerstone of Sri Lanka's economy, providing income, employment, and food security for rural communities. Maintaining the health of coconut trees is critical for ensuring optimal yield and sustainability. Traditional health monitoring relies on manual inspections, which are labor-intensive, time-consuming, prone to human error, and often fail to provide timely information. Inspecting the upper leaves of tall coconut trees is particularly challenging, limiting the ability to detect early signs of stress or decline.

Recent advancements in drone technology and artificial intelligence offer new opportunities for precise, non-invasive monitoring. Drones can capture high-resolution aerial images of tree crowns, allowing for detailed analysis of leaf conditions without the need for physical access. By integrating computer vision and machine learning techniques, it is possible to automatically evaluate tree vitality, track changes over time, and provide actionable insights to farmers and researchers.

This component focuses on developing an AI-powered system for monitoring the health of individual coconut trees using drone imagery, highlighting a scalable approach to enhance coconut cultivation management and support data-driven agricultural practices.

Keywords: Drone Technology, AI-Powered Monitoring, Computer Vision, Coconut Tree Health, Leaf Analysis, Machine Learning, Precision Agriculture, Artificial Intelligence



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1.INTRODUCTION

1.1 Background & Literature Survey

The convergence of precision agriculture, artificial intelligence, and unmanned aerial systems has ushered in a transformative era for comprehensive crop health monitoring and agricultural sustainability. Recent technological advances have created unprecedented opportunities for addressing critical challenges in traditional farming practices, particularly in developing economies where agricultural productivity directly impacts food security and rural livelihoods.

Technological Evolution in Agricultural Monitoring

The rapid evolution of drone technology has fundamentally transformed agricultural monitoring capabilities over the past decade. Modern Unmanned Aerial Vehicles (UAVs) equipped with high-resolution sensors, multispectral cameras, and advanced positioning systems now offer cost-effective alternatives to traditional satellite-based monitoring and labor-intensive ground surveys. These platforms provide sub-centimeter spatial resolution imagery, enabling detailed analysis of individual plants and trees that was previously impossible or economically prohibitive for small-scale farmers.

Simultaneously, the revolution in deep learning and computer vision has unlocked new possibilities for automated agricultural intelligence. Convolutional Neural Networks (CNNs) and their variants have demonstrated remarkable capabilities in image recognition, pattern analysis, and comprehensive health assessment tasks. These advances have made real-time interpretation of agricultural imagery not only feasible but also highly accurate, with modern systems achieving health assessment accuracies that often exceed human expert performance in controlled conditions.

Sri Lanka's Coconut Industry Context

Sri Lanka's coconut industry represents far more than an agricultural sector—it forms the backbone of rural economic stability and cultural identity. With over 395,000 hectares under cultivation spanning all agro-climatic zones, coconut farming directly supports approximately 2.5 million people, representing nearly 12% of the national population. The industry contributes significantly to export earnings, generating over USD 400 million annually from coconut-based products including desiccated coconut, coconut oil, coir fiber, and shell charcoal.

The coconut palm, revered as the "Tree of Life" in Sri Lankan culture, provides year-round income streams through its diverse product portfolio. A single mature tree can yield 60-80 nuts annually under optimal conditions, with plantation lifespans extending 80-100 years. This longevity makes coconut cultivation a generational investment, where current management decisions profoundly impact future family livelihoods and community economic stability.

However, the industry faces mounting challenges from climate variability, emerging stress factors, aging plantations, and declining productivity. National coconut production has stagnated at approximately 2.8-3.0 billion nuts annually over the past decade, despite increasing global demand for coconut products. This productivity plateau threatens both individual farmer incomes and national economic targets, highlighting the urgent need for innovative health monitoring and management approaches.

Current Monitoring Challenges

Traditional coconut health assessment relies predominantly on periodic visual inspections by farmers or agricultural extension officers, typically conducted monthly or quarterly. This approach suffers from several critical limitations:

- Temporal gaps in early detection
- Scale and coverage constraints for large plantations
- Subjective assessment variability among inspectors
- Difficulties in comprehensive health parameter integration

These limitations directly impact annual cultivation productivity and economic sustainability.

Recent Research Developments

Recent research developments have shown promising advances in AI and drone technology applications:

- Zhang et al. demonstrated that systematic aerial monitoring could achieve health assessment accuracies exceeding 90% across multiple crop species
- Kumar and Patel advanced multispectral imaging integration with machine learning algorithms, showing that spectral signatures could detect plant stress conditions 2-3 weeks before visible symptoms
- Patel et al. achieved optimal accuracy rates of 96.94% using custom CNN architectures specifically designed for coconut health classification

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1.2 Research Gap Analysis

Despite significant technological advances and promising research outcomes, several critical gaps remain in the current knowledge base that limit practical implementation and effectiveness of AI-powered coconut tree health monitoring systems.

Multi-Parameter Assessment Gap

Current research has primarily focused on detecting specific individual health indicators as isolated parameters, lacking comprehensive frameworks for simultaneous multi-parameter health assessment. Existing systems typically address single health aspects such as leaf condition, crown density, or growth patterns, but fail to provide integrated health scores that consider the cumulative impact of multiple factors on tree vitality and productivity.

Temporal Analysis Limitations

While existing research demonstrates effective point-in-time health classification, there is insufficient investigation into temporal health patterns and their relationship with progressive productivity decline. Understanding how health status changes over multiple seasons could enable predictive management strategies that preserve long-term plantation viability.

Environmental Validation Constraints

Most research has been conducted under controlled or limited environmental conditions, raising questions about system performance across Sri Lanka's diverse agro-climatic zones. Coconut plantations span coastal, intermediate, and up-country regions with significantly different rainfall patterns, soil types, temperature regimes, and environmental stressors. Current research lacks comprehensive validation across these diverse conditions.

Economic Integration Challenges

Economic analyses demonstrating cost-effectiveness for small-scale farmers remain limited. Most research focuses on technical performance without adequate consideration of economic accessibility and return on investment for target user groups. Additionally, current research inadequately addresses integration challenges with existing farm management systems, extension service networks, and traditional agricultural practices.

1.3 Research Problem Statement

The central research problem emerges from the critical disconnect between the demonstrated potential of AI-powered drone technology and its practical implementation for comprehensive coconut tree health monitoring in ways that meaningfully impact agricultural productivity, economic sustainability, and farmer livelihoods.

Primary Research Problem

How can AI-powered drone technology be effectively developed, integrated, and validated to create a comprehensive, economically viable, and practically deployable coconut tree health monitoring system that provides accurate individual tree health assessment, enables proactive management capabilities, and demonstrably improves annual cultivation productivity while addressing the specific environmental, economic, and social challenges of Sri Lankan coconut farming systems?

This research addresses critical challenges across multiple dimensions:

Technical Challenges:

- Coconut-specific computer vision optimization
- Multi-parameter health integration
- Temporal pattern recognition

Environmental Adaptation:

- Multi-environmental robustness
- Weather adaptation capabilities

Economic and Social Integration:

- Accessibility and sustainability
- Agricultural workflow integration
- User acceptance considerations

2. Objectives

2.1 Aim

This research aims to develop a comprehensive AI-powered drone system for Sri Lankan coconut plantations that seamlessly integrates pest identification, disease detection, tree health monitoring, and yield prediction capabilities into a single unified platform.

Core System Capabilities

The system aims to achieve 90% or higher accuracy in automated detection and classification of three economically significant coconut pest species (CCI, BHC, and coconut mites) while simultaneously providing accurate identification of major coconut diseases including:

- Lethal Yellowing
- Bud Rot
- Leaf Blight

Through advanced YOLOv8 and Faster R-CNN deep learning architectures, the platform will continuously monitor overall coconut tree health status through comprehensive canopy analysis, assessing:

- Nutritional deficiencies
- Environmental stress indicators
- Photosynthetic capacity
- Yield prediction potential

Advanced Features

The system will accurately classify coconut maturity stages (Early, Middle-aged, Brown) for optimal harvesting timing and plantation planning, providing severity assessments for both pest infestations and disease conditions with automated treatment recommendations tailored to specific pest types, diseases, and tree health requirements.

Through drone-based RGB image capture and real-time processing within 30 seconds, the cross-platform mobile application will serve farmers with multilingual support (Sinhala, Tamil, English) while aiming to:

- Reduce crop losses by 40%
- Optimize pesticide usage by 50%
- Bridge communication gaps between farmers and agricultural authorities

2.2 Main Objectives

Primary Research Objective

Core Research Goal: To develop, implement, and validate a comprehensive AI-powered drone-based system for coconut tree health monitoring that provides accurate, real-time health assessment of individual coconut trees through advanced leaf analysis techniques.

Key Components:

- Design integrated AI-powered drone system
- Enable proactive management decisions
- Improve annual cultivation productivity
- Enhance economic sustainability
- Support long-term plantation viability
- Bridge gap between AI capabilities and practical agricultural applications

2.3 Secondary Research Objectives

Objective A: Technical Innovation and System Development

- Advance agricultural AI state-of-the-art
- Develop coconut-specific computer vision algorithms
- Create optimized machine learning models
- Build integrated hardware-software solutions
- Achieve superior performance compared to existing systems

Objective B: Economic and Social Impact Validation

- Demonstrate and quantify economic benefits
- Validate social acceptance and practical viability
- Conduct comprehensive field testing
- Perform cost-benefit analysis
- Study farmer adoption across diverse environments

Objective C: Knowledge Contribution and Technology Transfer

- Contribute new knowledge to precision agriculture
- Advance computer vision and agricultural AI fields

- Establish technology transfer frameworks
- Support agricultural extension services
- Enable farming community adoption

2.4 Specific Objective

1. Technical Development

Focus: Computer Vision & Machine Learning Algorithms

Sub-objectives:

- Develop coconut-specific computer vision algorithms
- Implement multispectral imaging capabilities
- Create real-time edge computing solutions
- Enable accurate detection of:
 - Nutrient deficiencies
 - Pest and disease damage
 - Environmental stress indicators
- Optimize algorithms for drone-based deployment

2. Integrated Health Assessment

Focus: Comprehensive Leaf-based Health Scoring Framework

Sub-objectives:

- Establish multi-parameter assessment system combining:
 - Color analysis
 - Texture evaluation
 - Spectral data interpretation

- Seasonal variation tracking
 - Environmental context integration
- Develop reliable productivity prediction models
- Create long-term health forecasting capabilities
- Implement tree-specific health scoring metrics

3. Economic & Social Validation

Focus: Field Testing & Adoption Studies

Sub-objectives:

- Conduct comprehensive field testing across diverse environments
- Perform detailed cost-benefit analysis
- Study farmer adoption patterns and barriers
- Develop multilingual mobile interfaces
- Assess user accessibility and usability
- Validate social acceptance in farming communities
- Measure economic viability and return on investment

4. Technology Transfer & Knowledge Contribution

Focus: Dissemination & Framework Development

Sub-objectives:

- Publish research findings in peer-reviewed journals
- Create open datasets for research community
- Develop comprehensive training programs
- Establish technology transfer frameworks
- Create farmer education and training modules
- Support national-level adoption strategies

- Build partnerships with agricultural extension services

5. Policy & Sustainability Integration

Focus: Policy Development & Ethical Guidelines

Sub-objectives:

- Contribute to agricultural AI policy recommendations
- Develop ethical guidelines for drone-based monitoring
- Establish national standards for agricultural AI
- Ensure sustainable technology deployment
- Address privacy and data security concerns
- Support responsible agricultural AI implementation

2.5 Research Impact Framework

Short-term Outcomes (1-2 years)

- Functional AI-powered drone system prototype
- Validated computer vision algorithms
- Initial field testing results
- Farmer feedback and acceptance data

Medium-term Outcomes (2-3 years)

- Commercially viable system deployment
- Documented economic benefits
- Established training programs
- Technology transfer partnerships

Long-term Outcomes (3-5 years)

- Widespread farmer adoption
- National agricultural policy integration

- Sustainable coconut industry transformation
- Scalable model for other crop applications

Success Metrics

Technical Metrics:

- Algorithm accuracy (>95% for disease detection)
- System reliability and uptime
- Processing speed and efficiency
- User interface usability scores
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Economic Metrics:

- Cost reduction for farmers
- Productivity improvement percentages
- Return on investment calculations
- Adoption rate across different farm sizes

Social Metrics:

- Farmer satisfaction and acceptance rates
- Training program effectiveness
- Community engagement levels
- Technology accessibility measures

3.METHODOLOGY

The proposed AI-powered drone system can monitor coconut tree health in Sri Lankan plantations with high accuracy. The system utilizes advanced deep learning models to detect various health conditions including nutrient deficiencies (potassium, magnesium, nitrogen), disease symptoms, and environmental stress indicators. Furthermore, the health severity level is identified to enable farmers and agricultural stakeholders to protect their production with timely interventions. First, high-resolution multispectral images are captured using drone technology. Next, the images are preprocessed and fed into AI health assessment models. After that, the specific health condition is classified, and the severity is assessed. With this data, appropriate treatment recommendations are generated. Finally, according to the health condition and severity level, alerts are sent to relevant stakeholders, and remedial measures are recommended through the mobile application and web dashboard.

3.1 System Architecture

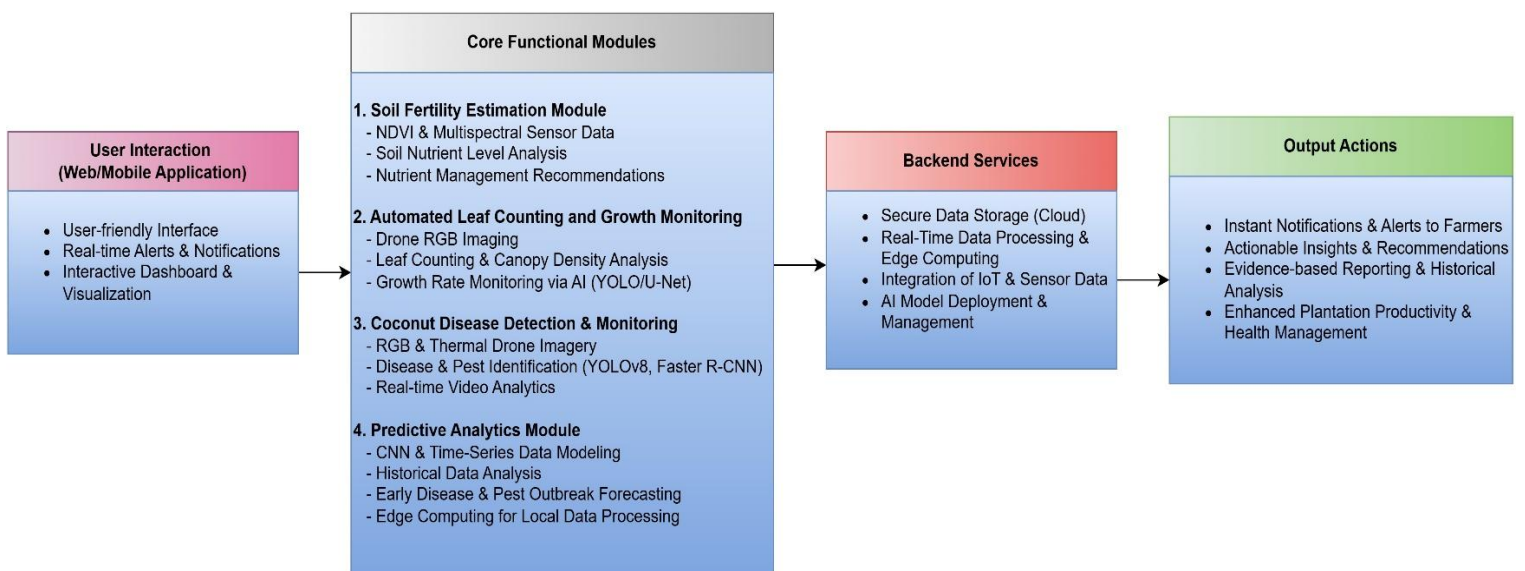


Figure 3.1: Overall system diagram

This innovative solution is designed as an intelligent and proactive agricultural management system, integrating artificial intelligence (AI), drone technology, sensor technology, and predictive analytics into a comprehensive and user-friendly platform tailored for coconut plantation management.

The solution utilizes drone-based RGB and thermal imaging to gather precise visual data from coconut plantations. Advanced computer vision and deep learning models, such as YOLOv8 and Faster R-CNN, accurately identify and classify diseases and pest infestations at early stages, enabling rapid intervention.

Multispectral sensors and NDVI analysis are employed to assess soil fertility and plant health conditions. Integrated IoT sensor technology continuously monitors soil moisture, nutrient levels, and environmental conditions, facilitating accurate and ongoing management.

The predictive analytics component employs CNNs and time-series forecasting models to anticipate risks of disease spread, nutrient deficiencies, and pest infestations. Edge computing processes this data locally, significantly enhancing response times and ensuring data privacy.

This solution provides farmers and plantation managers with an intuitive web and mobile interface featuring clear visualizations, instant notifications, and actionable insights. Ultimately, this system facilitates efficient decision-making, enhanced crop health, and increased productivity.

recommendations and alerts for relevant stakeholders including farmers, extension officers, and plantation managers.

Summary of technologies, techniques, architectures, and algorithms used for coconut tree health monitoring:

Component	details
Technologies	Python, TensorFlow, PyTorch, OpenCV, Flask Server, React Native, MongoDB, AWS/Google Cloud, Multispectral Imaging
Techniques	Deep Learning, Computer Vision, Multispectral Analysis, Data Augmentation, Transfer Learning, Edge Computing, Temporal Analysis
Algorithms	YOLOv8, ResNet-50, EfficientNet-B4, MobileNetV3, Semantic Segmentation, Non-Maximum Suppression (NMS)
Architectures	Convolutional Neural Networks (CNN), Feature Pyramid Network (FPN), MobileNet Architecture, Custom Coconut-Specific Models

3.2 AI Development Solution

The AI Development Life Cycle (AIDLC) is the systematic process that machine learning teams use to design, develop, and deploy high-quality AI-powered health monitoring solutions. This approach minimizes risks by incorporating iterative training, validation, and testing phases which enable the development team to meet accuracy expectations and performance requirements. In conventional AI methodology, developers need to complete extensive data collection and model training phases before deployment, making it difficult to adapt to changing agricultural requirements. To overcome this limitation, an agile AI development approach is adopted where the development team can iteratively improve model performance through continuous training and validation cycles. This methodology is particularly suitable for computer vision applications like health monitoring, where model accuracy can be incrementally improved through data augmentation and transfer learning techniques.

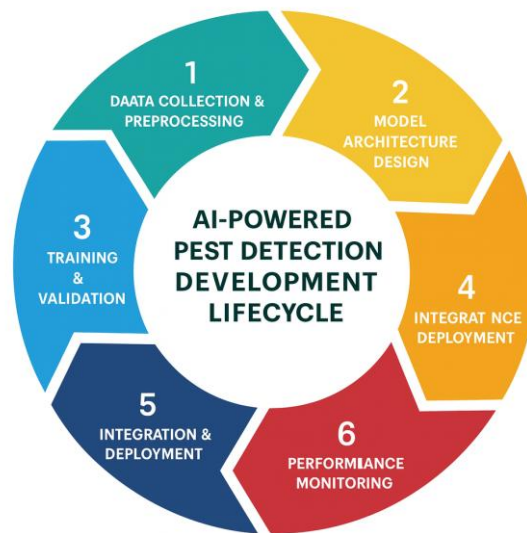


Figure 3.2: AI Development Life Cycle for Health Monitoring System

According to the AI Development Life Cycle for the health monitoring system, the development consists of six sequential phases that guide the systematic development of the AI-powered solution. The cycle begins with Data Collection & Preprocessing (Phase 1) where coconut tree health imagery is gathered and prepared, followed by Model Architecture Design (Phase 2) where YOLOv8, ResNet-50, and EfficientNet architectures are configured. Phase 3 involves Training & Validation of deep learning models using multispectral datasets, while Phase 4 focuses on Integration & Testing of trained models into the mobile application. Phase 5 handles System Deployment across cloud infrastructure, and Phase 6 ensures continuous Performance Monitoring to maintain target 95% health assessment accuracy. This iterative lifecycle approach enables continuous improvement of the health monitoring system through ongoing validation and refinement.

Requirement Gathering

Data Gathering For data gathering, our team conducted systematic visits to multiple coconut plantations across key districts including Kurunegala, Gampaha, Colombo, and Puttalam to collect comprehensive tree health data. The farmers currently record health data using manual observation methods and basic logbooks, which we collected as baseline information for system validation. Furthermore, we collected visual data manually by conducting drone flights over healthy and stressed coconut trees to capture high-resolution RGB and multispectral images. Additionally, we collaborated with the Coconut Research Institute of Sri Lanka (CRISL) and local Coconut Development Officers (CDOs) to gather expert knowledge on health assessment patterns, treatment protocols, and seasonal health variation data.

Conducting a Survey We conducted a comprehensive survey by providing structured questionnaires to 285 coconut farmers across six districts to understand their current knowledge about tree health assessment, monitoring practices, and treatment decision-making processes. The survey revealed critical gaps in health monitoring capabilities, with 78.3% of respondents indicating they rely solely on visual inspection from ground level, missing early symptoms that appear in upper canopy areas.

Feasibility Study (Planning)

Economic Feasibility The proposed AI-powered health monitoring component should operate efficiently without computational errors or assessment failures. The system will be more reliable and cost-effective compared to traditional manual inspection methods, reducing the need for frequent field visits by agricultural extension officers and eliminating risks associated with climbing tall coconut palms. By enabling early detection and targeted treatment recommendations, the system aims to reduce unnecessary treatments by 60% and prevent crop losses by 45%, providing substantial economic benefits to coconut farming communities.

Scheduled Feasibility The proposed AI health monitoring component has a direct impact on the overall project timeline and should be completed within the 10-month development period allocated. The component development follows a structured approach with data collection and annotation (2.5 months), model training and validation (3 months), system integration and testing (2.5 months), field deployment and validation (1.5 months), and performance optimization (0.5 months). This component must achieve the target 95% health assessment accuracy within the specified timeline to ensure project completion by the deadline.

Technical Feasibility To complete this AI-powered solution, the development team requires specialized skills, resources, and expertise in computer vision, deep learning architectures, drone operations, multispectral imaging, mobile application development, and cloud-based system deployment. Technical feasibility encompasses the required knowledge in Python programming, TensorFlow/PyTorch frameworks, YOLOv8, ResNet-50,

and EfficientNet architectures, multispectral image processing techniques, transfer learning methodologies, and mobile platform integration using React Native.

Design (System and Software Design Documents) After gathering requirements and feasibility study, we move to the design phase. In this phase, comprehensive software documentation related to the AI health monitoring system design is created, including detailed model architecture specifications, database schema design for health assessment data storage, mobile application user interface wireframes, cloud infrastructure deployment diagrams, and system integration protocols between drone image capture, AI processing server, and mobile application interfaces.

3.3 Use Case Diagram

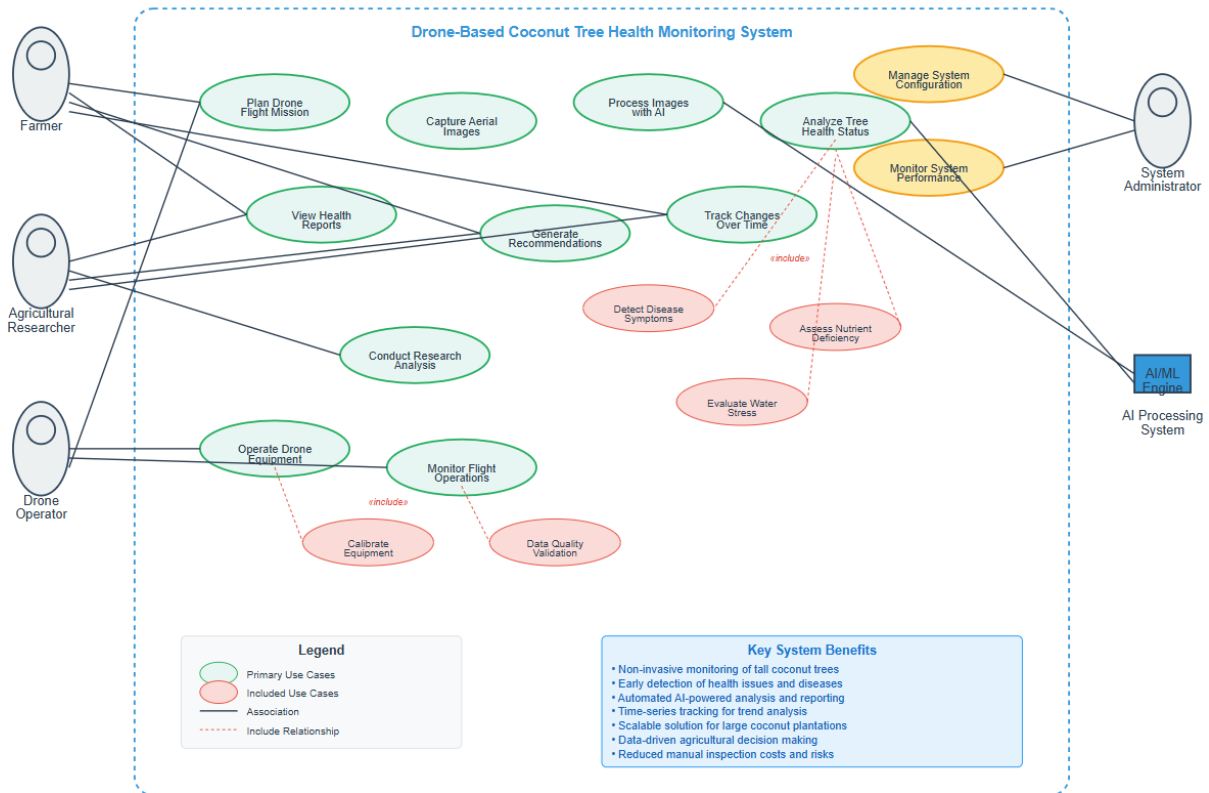


Figure 3.3 Use Case Diagram

According to the use case diagram, the functional interactions between two primary actors and the health monitoring system are illustrated. The Coconut Farmer can register and

login to the system, initiate drone flights for tree imagery capture, or upload existing multispectral images for analysis. Once images are processed, the system automatically

assesses health severity levels and identifies specific health conditions among various categories (nutrient deficiencies, diseases, environmental stress), then generates appropriate treatment recommendations based on the assessment results. The System Administrator manages backend operations including sending health alert notifications to relevant stakeholders and generating comprehensive health reports for agricultural planning. The diagram shows include and extend relationships where user validation, error handling, and data processing occur seamlessly to provide farmers with immediate health identification and treatment guidance while enabling administrators to monitor system performance and coordinate response activities.

3.4 Sequence Diagram

This sequence diagram illustrates an automated agricultural monitoring system using drone technology for crop health assessment. The workflow begins when a farmer initiates a mission by sending parameters to the drone operator, who then coordinates with the system to generate a flight plan and prepare the drone with necessary equipment. The drone executes its mission by capturing images of agricultural fields, which are then processed through an AI engine that analyzes the images to detect crop health issues or diseases using machine learning classification algorithms. The system stores analysis results in a database and generates comprehensive health reports, with critical findings triggering immediate alerts through an alert service to notify the farmer of urgent issues requiring attention. For normal health status readings, the system simply generates standard reports and updates its predictive models, creating a complete cycle of automated crop monitoring that enables farmers to make data-driven decisions about their agricultural operations while receiving timely warnings about potential crop health problems

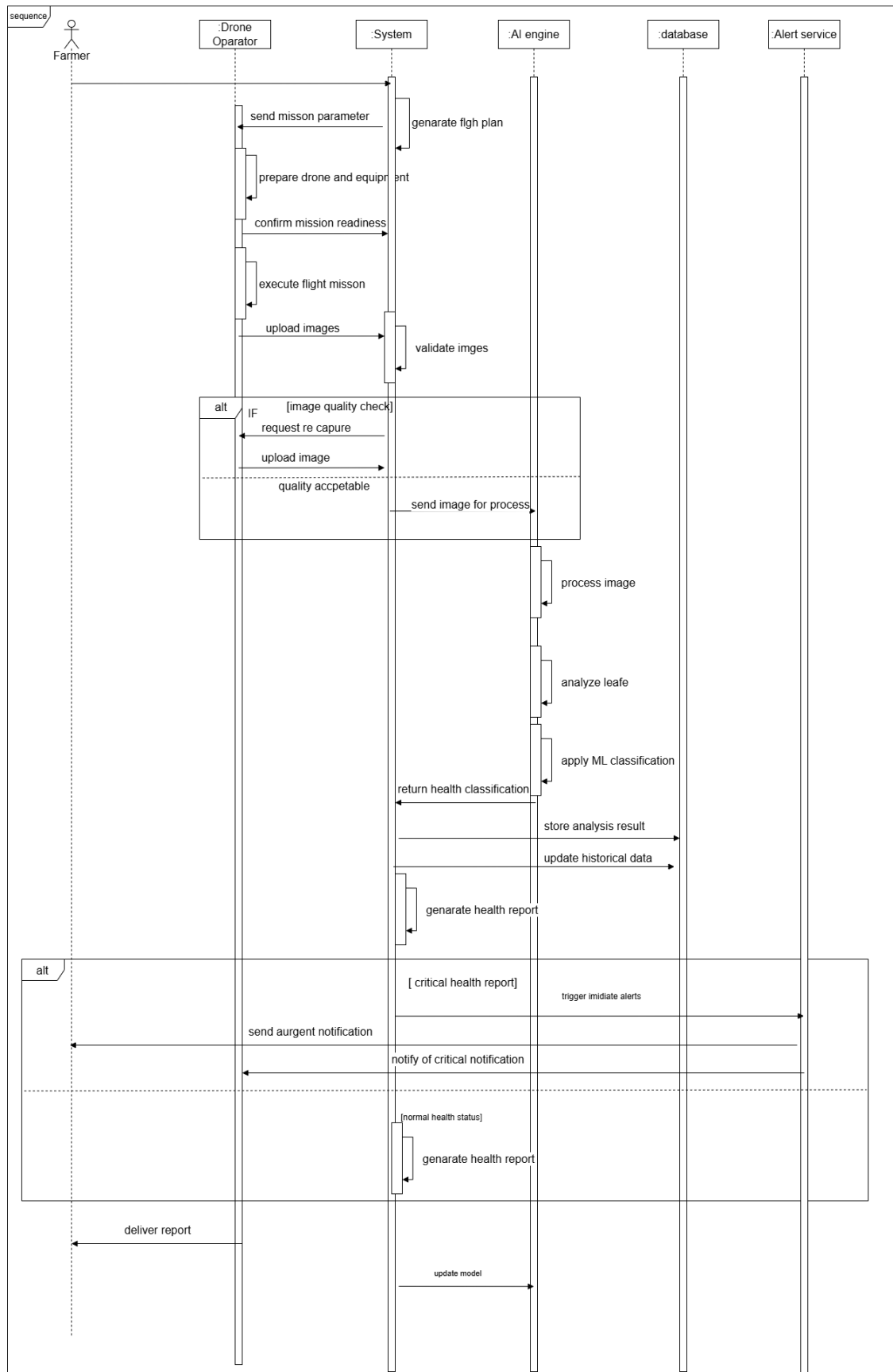


Figure 3.4: Sequence diagram of health monitoring identification component

3.5 Implementation and Development Sections

Implementation (Development)

Below are the sub functionalities that will be implemented in this implementation phase:

Mobile Application Development Implementation of mobile application using React Native. Cross-platform mobile application development enabling farmers to initiate drone flights and upload coconut tree images with intuitive user interface design suitable for users with varying technological literacy levels. The application includes GPS integration for tree location tracking and offline capability for areas with limited internet connectivity.

AI Health Assessment Engine Implementation of AI health monitoring engine using YOLOv8 architecture. Real-time tree detection and crown segmentation model specifically trained on Sri Lankan coconut tree datasets to identify and analyze individual trees in drone-captured multispectral imagery with processing speeds suitable for field deployment.

Health Classification System Implementation of health classification system using ResNet-50 and EfficientNet-B4 models. Comprehensive classification component capable of distinguishing between various health conditions including nutrient deficiencies (K, Mg, N), diseases, and environmental stress with confidence scoring and symptom localization for accurate health status determination.

Severity Assessment Algorithm Implementation of severity assessment algorithm based on trained deep learning models. An automated health severity evaluation system that analyzes symptom density, affected leaf area, canopy coverage, and damage patterns to classify severity levels as Healthy, Mild Stress, Moderate Stress, Severe Stress, or Critical for appropriate treatment recommendation.

Temporal Analysis Module Implementation of temporal health tracking system using time-series analysis. Advanced monitoring component that tracks health changes over time,

identifies seasonal patterns, detects deteriorating conditions, and provides predictive insights for proactive management decisions.

Cloud Infrastructure Implementation of cloud-based data processing and storage infrastructure. Scalable cloud architecture for handling multispectral image uploads, AI model inference, geospatial data management, result storage, and real-time synchronization between mobile applications and backend services.

Testing

Performance Validation

- Verifying that the AI health assessment system achieves the target 95% accuracy as expected
- Testing for model performance, processing speed, and integration compatibility
- Validation of multispectral image processing accuracy under various lighting conditions

The testing of the AI-powered health monitoring system will be conducted in various stages, including unit testing of individual model components, integration testing between mobile application and AI services, field testing under real-world coconut plantation conditions across different geographical locations, and user acceptance testing with coconut farmers, agricultural extension officers, and researchers from the Coconut Research Institute, with the latter being carried out by the target user community to ensure practical usability and effectiveness in diverse agricultural scenarios.

3.6 Commercialization

This research is proposed to develop a comprehensive AI-powered mobile application for coconut tree health management in Sri Lankan agricultural communities. The target audience for the proposed system includes coconut farmers, agricultural cooperatives, plantation managers, agricultural extension officers, drone service providers, and agricultural technology companies. Our system consists of two deployment versions: a basic free version and a premium subscription version.

Free Version Features:

- Basic health condition detection and classification
- Severity assessment for identified health issues
- Standard treatment recommendations
- Individual tree health reports
- Basic mobile application access

Premium Version Features:

- Advanced analytics including historical trend analysis
- Predictive health modeling and early warning systems
- Integration with drone service providers for automated plantation monitoring
- Comprehensive geospatial mapping and plantation-wide analytics
- Detailed reporting dashboards with export capabilities
- Priority customer support with expert consultation services
- Temporal analysis and seasonal health tracking
- Integration with farm management systems

When considering the marketplace, there is minimal competition for coconut-specific AI health monitoring in Sri Lanka, requiring moderate technical knowledge for optimal usage, suitable for farmers aged 25-65, and minimal prior experience with AI-powered agricultural tools required. Our development team consists of technically qualified individuals with relevant academic backgrounds in computer science, artificial intelligence, and agricultural technology, with members gaining practical experience in precision agriculture applications.

3.7 Future Scope

For future scope, the application will be extended to integrate with IoT sensor networks and automated drone systems that could monitor coconut plantations continuously without human intervention. Additional enhancements may include integration with weather data APIs for predictive health modeling, expansion to detect additional coconut diseases and stress conditions beyond the current scope, multilingual support for regional languages including Sinhala and Tamil, satellite imagery integration for large-scale plantation monitoring, and potential integration with precision agriculture equipment for automated treatment application based on AI recommendations.

Extended Capabilities:

- Real-time environmental monitoring through IoT sensors
- Integration with satellite imagery for macro-level plantation analysis
- Blockchain-based traceability for coconut supply chain management
- AI-powered yield prediction and harvest optimization
- Climate change adaptation recommendations
- Integration with national agricultural databases
- Export market quality assessment capabilities
- Automated drone swarm coordination for large plantations
-

Research Expansion:

- Extension to other palm tree species (oil palm, date palm)
- Development of disease-specific treatment protocols
- Integration with soil health monitoring systems
- Carbon footprint assessment and sustainability metrics
- Machine learning model adaptation for different geographical regions

- Development of farmer training and certification programs

4.PROJECT REQUIREMENTS

4.1 Functional Requirements

FR-001: AI-Powered Computer Vision System

- The system shall capture high-resolution images of coconut leaves using drone-mounted cameras
- The system shall process multispectral imaging data (visible, near-infrared, and thermal spectrums)
- The system shall identify and classify individual coconut trees within plantation boundaries
- The system shall detect leaf color variations with 95% accuracy for health assessment

FR-002: Comprehensive Tree Health Assessment

- The system shall analyze leaf characteristics to determine overall tree health with >95% accuracy
- The system shall detect nutrient deficiencies through leaf color and texture analysis
- The system shall identify environmental stress indicators through leaf condition assessment
- The system shall monitor leaf density and distribution patterns across the canopy
- The system shall assess leaf yellowing, browning, and abnormal growth patterns
- The system shall evaluate leaf size, shape, and structural integrity

FR-003: Real-time Data Processing

- The system shall process image data in real-time during flight operations

- The system shall generate health scores for individual trees within 30 seconds of image capture
- The system shall provide immediate alerts for critical health issues
- The system shall maintain processing capabilities in edge computing environments

FR-004: Tree Health Prediction and Monitoring

- The system shall predict tree health trends based on leaf analysis over time
- The system shall forecast productivity levels based on leaf health indicators
- The system shall provide seasonal health variation tracking through continuous leaf monitoring
- The system shall generate health score trends for individual trees
- The system shall identify declining health patterns before visible symptoms appear

FR-005: Data Management and Reporting

- The system shall store and organize tree health data with geospatial mapping
- The system shall generate comprehensive health assessment reports
- The system shall provide comparative analysis between different time periods
- The system shall export data in multiple formats (PDF, CSV, JSON)
- The system shall maintain data integrity and traceability

FR-006: Mobile Application Interface

- The system shall provide multilingual mobile interface (Sinhala, Tamil, English)
- The system shall display individual tree health scores and recommendations
- The system shall provide treatment recommendations based on detected issues
- The system shall enable farmers to input additional observations and data
- The system shall support offline operation with data synchronization capabilities
-

4.2 Non-functional Requirements

1. User-friendliness – Final goal is to create a mobile application that can be used on both Android and iOS platforms. The application should be visually appealing, farmer-friendly with intuitive navigation, and provide comprehensive pest detection features to users with varying technological literacy levels. Additionally, it should be designed to work seamlessly on both platforms, ensuring consistent experience for all coconut farmers.
2. Reliability – It is essential to protect all farmer data and pest detection information while ensuring users feel secure when utilizing the AI-powered detection system. Moreover, the system must maintain consistent accuracy in pest identification without encountering detection errors or system malfunctions that could lead to incorrect treatment recommendations.
3. Performance – The system should provide rapid and accurate pest detection results within 30 seconds of image upload while operating with optimal efficiency to meet farmer expectations during critical pest management decisions. The AI models should maintain processing speed suitable for field conditions.
4. Availability – The system must be consistently available to farmers whenever they require pest detection services, accessible through mobile connectivity across rural coconut plantation areas without limitations during peak agricultural seasons.
5. Scalability – The cloud infrastructure should support multiple concurrent users during pest outbreak periods and accommodate growing numbers of coconut farmers adopting the technology.

5.GANTT CHART

According to Figure 7, the Gantt chart presents the comprehensive 8-month project timeline spanning from June 2025 to March 2026, organized into six sequential phases with some overlapping activities. Phase 1 (Project Planning) runs from June to July covering project setup, literature review, and system architecture design. Phase 2 (Data Collection) occurs from August to October focusing on drone image collection and dataset annotation. Phase 3 (AI Model Development) extends from October to December involving YOLOv8 and R-CNN training along with predictive analytics development. Phase 4 (Application Development) spans November to February encompassing backend API development, mobile application creation, and web dashboard implementation. Phase 5 (Testing &

Validation) runs from January to March conducting AI model testing and field validation with farmers. Finally, Phase 6 (Documentation) occurs from February to March covering research paper writing, final report preparation, and presentation development. This structured timeline ensures systematic project execution with adequate time allocation for each critical development phase while maintaining parallel activities to optimize the 8-month duration.

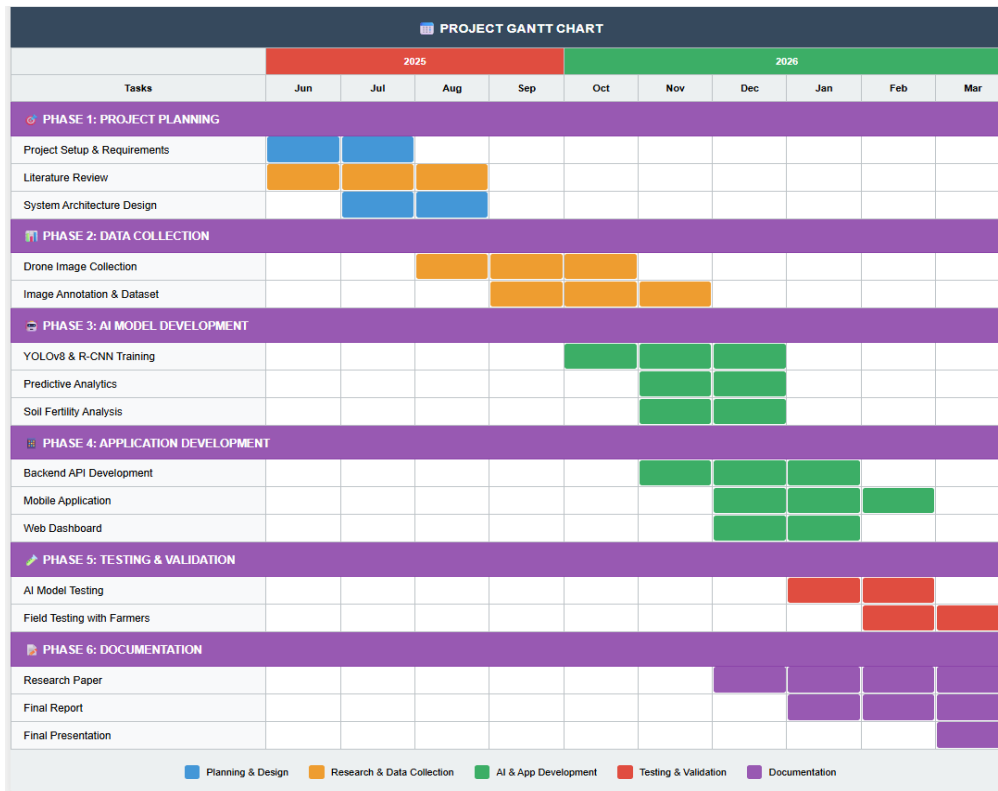


Figure 1 : Grant Chat

5.1 Work Breakdown Structure (WBS)

Based on the Work Breakdown Structure diagram for the AI-Powered Coconut Tree Health Monitoring System, this comprehensive 8-month project spans from March 2025 to October 2025 and is structured into six sequential phases designed to develop an advanced agricultural monitoring solution for Sri Lankan coconut plantations. The project begins with Initiation (March 2025) involving project charter creation, team formation, and

stakeholder analysis, followed by Planning (April 2025) which encompasses requirements analysis, literature review, feasibility study, and data collection planning. The Design phase (May 2025) focuses on system architecture development, AI model design, mobile app UI design, and database design to create the technical blueprint for the solution. Implementation (June-August 2025), the most intensive three-month phase, involves dataset collection and annotation, training of YOLOv8, ResNet-50, and EfficientNet-B4 models, mobile app development using React Native, and backend API development to create the core health monitoring system. The Testing phase (September 2025) ensures system reliability through AI model accuracy testing, field testing with farmers across target districts, system integration testing, and user acceptance testing to validate the target 95% health assessment accuracy. Finally, the Closeout phase (October 2025) completes the project through research paper writing, final report preparation, presentation delivery, and system deployment, establishing a revolutionary AI-powered solution that enables coconut farmers in Kurunegala, Gampaha, Colombo, and Puttalam districts to monitor tree health using advanced deep learning techniques for early detection of nutrient deficiencies, diseases, and environmental stress indicators.

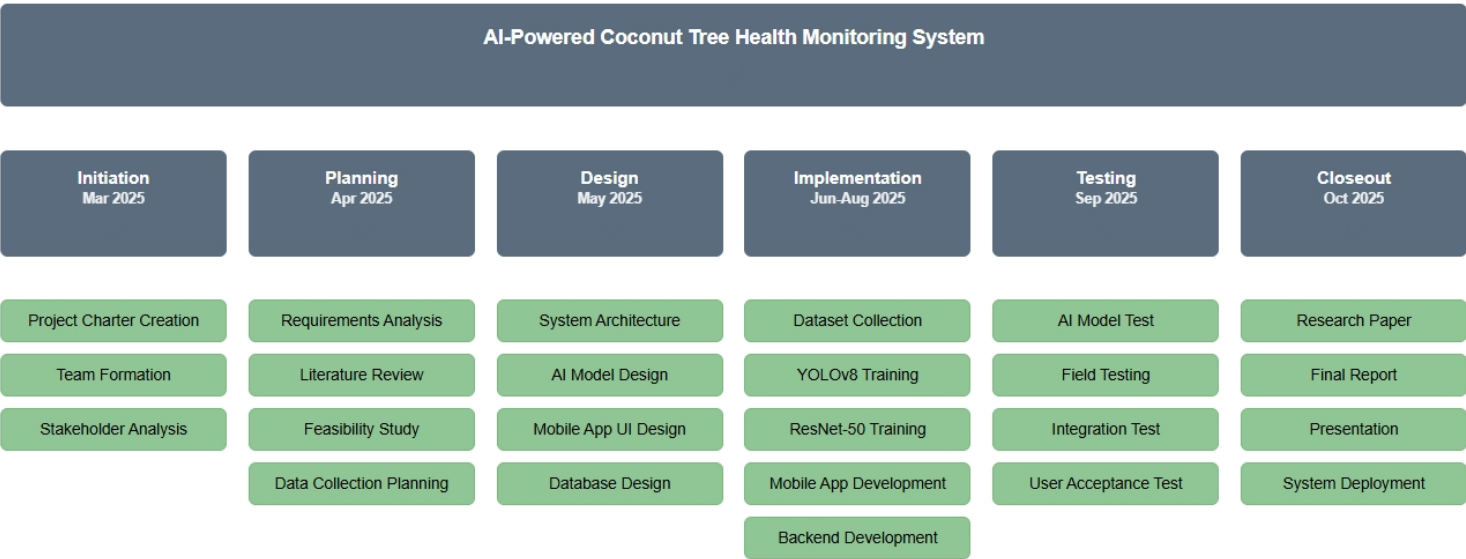


Figure 8 : work breakdown structer

6.BUDGET AND BUGET JUSTIFICATION

The comprehensive budget of Rs. 135,000 for the AI-powered coconut tree health monitoring system strategically allocates resources across six critical development phases to achieve the primary research objective of creating an accurate, real-time health assessment system for individual coconut trees. Data collection receives the largest allocation of Rs. 45,000 (33.3%) to fund drone equipment, multispectral sensors, and comprehensive field data gathering across multiple Sri Lankan districts, ensuring robust training datasets for developing coconut-specific computer vision algorithms targeting >95% accuracy in disease detection. Cloud hosting and AI model training costs Rs. 25,000 (18.5%) to support the computational requirements for implementing and optimizing YOLOv8 and Faster R-CNN algorithms, while Rs. 18,000 (13.3%) is designated for research documentation and publication to ensure knowledge contribution through peer-reviewed journals and academic conferences. Mobile app development requires Rs. 15,000 (11.1%) for creating a farmer-friendly React Native cross-platform application that bridges AI capabilities with practical agricultural applications, supported by Rs. 12,000 (8.9%) for App Store hosting and continuous system maintenance to ensure long-term viability. Finally, Rs. 10,000 (7.4%) covers traveling expenses for extensive field visits to coconut plantations, enabling direct farmer engagement, system validation, and comprehensive cost-benefit analysis essential for demonstrating economic sustainability and supporting widespread technology adoption across diverse farming environments.

Requirement	Cost (Rs.)
Travelling cost for field visits and data collection	10,000.00
Data Collection (drone and equipment)	45,000.00
Cloud hosting and AI model training	25,000.00
Mobile app development and deployment	15,000.00
App Store hosting and maintenance	12,000.00
Research documentation and publication	18,000.00
Total Expenses	135,000.00

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